

Scan2Fab AI | System Architecture

3-Phase Technical Pipeline for MSME Reverse Engineering

Highly Scalable

GPU Cloud Native

PHASE 1: INGESTION

Physical to Digital

Capturing the physical object geometry entirely without laser hardware.



Standard Smartphone Camera

Worker records a simple 360-degree orbital 4K video of the broken part.



Encrypted Cloud Uplink

2D video frames are extracted, encrypted, and streamed to the high-performance AWS/GCP GPU cluster.

PHASE 2: COMPUTATION

AI Neural Engine

The core frontier technology utilizing Generative Spatial AI.



Neural Radiance Fields (NeRF)

Deep neural networks learn the volumetric scene representation, calculating exact depth from flat 2D pixels.



3D Gaussian Splatting

Ultra-fast rendering generates a high-density, dimensionally accurate point cloud of the mechanical part.



Mesh Synthesis

Point cloud is converted into a solid, watertight polygonal mesh, sealing any optical holes automatically.

PHASE 3: FABRICATION

Digital to Physical

Returning the digital twin back into the real world as a finished product.



Universal CAD Export

System instantly outputs standardized .STEP or .STL files, bypassing manual AutoCAD drafting completely.



Factory CNC Execution

File is fed directly into MSME CNC machines or Industrial 3D Printers to fabricate the replacement part.